EA-0021 - AQELYN Engineering Archive

AQELYN - EA-0021 Engineering Archive

IS-021 - AQELYN Predictive Analytics & Forecasting Engine

Archive ID: EA-0021

Implementation Specification: IS-021

Component: AQELYN Predictive Analytics & Forecasting Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Engineering Phase: Phase 3

Predecessor Archives: EA-0001 through EA-0020

Next Specification: IS-022 - AQELYN Executive Intelligence & Strategic Reporting Engine

Document Control

Field	Value
Document	Engineering Archive EA-0021
Specification	IS-021 - AQELYN Predictive Analytics & Forecasting Engine
Publication Format	Markdown, PDF, HTML, ZIP
Source of Truth	MD/EA-0021.md
Archive Rule	Implementation Specification -> Engineering Archive -> Continue
Repository Rule	Fixed repository structure; no redesign
Completion State	IS-021 complete; EA-0021 generated

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

Engineering Archive	Implementation Specification	Component
EA-0001	IS-001	AQELYN Kernel
EA-0002	IS-002	Universal Object Model
EA-0003	IS-003	AQELYN Event Bus

| EA-0004 | IS-004 | AQELYN Evidence Engine |

| EA-0005 | IS-005 | AQELYN Knowledge Graph |

| EA-0006 | IS-006 | AQELYN Trust Engine |

| EA-0007 | IS-007 | AQELYN Mission Engine |

| EA-0008 | IS-008 | AQELYN Workflow Engine |

| EA-0009 | IS-009 | AQELYN Policy Engine |

| EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |

| EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |

| EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |

| EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |

| EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |

| EA-0015 | IS-015 | AQELYN Security Operations (SOC) Engine |

| EA-0016 | IS-016 | AQELYN Digital Forensics Engine |

| EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |

| EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |

| EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |

| EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |

| EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |

The fixed repository structure remains:

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification
↓
Generate Engineering Archive
↓
Continue
```

No Engineering Archive may be skipped.

2. IS-021 Specification Identity

```
Specification ID: IS-021
Name: AQELYN Predictive Analytics & Forecasting Engine
Engineering Archive Target: EA-0021
Project: AQELYN
System Type: Cyber Security Operating Environment
Status: Complete
Predecessor: IS-020 - AQELYN AI Decision Intelligence Engine
```

3. Purpose

The AQELYN Predictive Analytics & Forecasting Engine provides advanced predictive analysis capabilities across the AQELYN platform by forecasting cyber threats, operational risks, mission impacts, infrastructure failures, and security trends using historical telemetry, AI decision intelligence, threat intelligence, and knowledge graph relationships.

The engine enables proactive rather than reactive cybersecurity operations.

It answers:

```
What is likely to happen next?
Which assets are at greatest future risk?
Which threats are most likely to evolve?
What incidents are likely to escalate?
Which missions require preventive action?
What trends indicate emerging attacks?
How confident are the forecasts?
Can forecasts be explained and audited?
```

4. Mission

The engine shall provide:

```
Predictive analytics
Threat forecasting
Risk forecasting
Mission forecasting
Behavior prediction
Trend analysis
Anomaly forecasting
Capacity forecasting
Operational forecasting
Confidence estimation
Explainable predictions
Continuous model improvement
```

5. Scope

5.1 In Scope

```
Threat prediction
Incident forecasting
Risk forecasting
Mission impact prediction
Trend analysis
Behavior forecasting
Security posture prediction
Infrastructure forecasting
Historical pattern analysis
Predictive scoring
Executive forecasting dashboards
```

5.2 Out of Scope

```
Financial forecasting
Business sales prediction
Consumer AI assistants
Autonomous offensive actions
Non-security business analytics
```

6. Dependencies

IS-021 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
IS-018 Automated Response & Orchestration Engine
IS-019 Security Data Lake & Telemetry Platform
IS-020 AI Decision Intelligence Engine
```

7. High-Level Architecture

```
AQELYN Predictive Analytics & Forecasting Engine
■
■■■ Forecast Engine
■■■ Trend Analysis Engine
■■■ Prediction Engine
■■■ Confidence Engine
■■■ Explainability Engine
■■■ Simulation Engine
■■■ Scenario Engine
■■■ Knowledge Graph Connector
■■■ Data Lake Connector
■■■ AI Decision Connector
■■■ Risk Connector
■■■ Mission Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-021-001 - Predictive Forecasting

The engine shall forecast:

```
Threat evolution
Incident escalation
Risk exposure
Mission disruption
Infrastructure degradation
Security posture
```

FR-021-002 - Trend Analysis

Support analysis of:

```
Historical telemetry
Threat trends
Attack campaigns
Risk trends
Operational metrics
Mission performance
```

FR-021-003 - Scenario Simulation

Support:

```
What-if analysis
Mission simulation
Threat propagation
Response effectiveness
Recovery scenarios
Risk reduction analysis
```

FR-021-004 - Confidence Assessment

Calculate:

```
Forecast confidence
Prediction confidence
Scenario confidence
Trend confidence
Model confidence
```

FR-021-005 - Forecast Governance

Support:

```
Human review
Policy validation
Auditability
Forecast traceability
Version control
```

FR-021-006 - Event Publication

Publish standardized events:

```
forecast.generated
forecast.updated
trend.detected
prediction.completed
scenario.simulated
confidence.updated
```

9. Non-Functional Requirements

The engine shall provide:

```
Explainability
High scalability
Low-latency forecasting
Continuous operation
Auditability
Repository stability
Backward compatibility
```

10. Core Forecast Lifecycle

```
Historical Data
↓
Trend Analysis
↓
Prediction
↓
Scenario Simulation
↓
Confidence Assessment
↓
Human Review
↓
Forecast Publication
```

11. Internal Component Architecture

The AQELYN Predictive Analytics & Forecasting Engine is implemented as a modular predictive analytics platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, AI Decision Intelligence Engine, Risk Intelligence Engine, and Mission Engine.

```
AQELYN Predictive Analytics & Forecasting Engine
■
■■■ Forecast Engine
■■■ Prediction Engine
■■■ Trend Analysis Engine
■■■ Simulation Engine
■■■ Scenario Engine
■■■ Confidence Engine
■■■ Explainability Engine
■■■ Learning Engine
■■■ Knowledge Graph Connector
■■■ Data Lake Connector
■■■ AI Decision Connector
■■■ Risk Connector
■■■ Mission Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Forecast Engine

Generates predictive forecasts.

Capabilities:

```
Threat forecasting
Risk forecasting
Mission forecasting
Incident forecasting
Infrastructure forecasting
```

12.2 Prediction Engine

Produces predictive models.

Functions:

```
Behavior prediction
Threat prediction
Risk prediction
Operational prediction
Mission prediction
```

12.3 Trend Analysis Engine

Analyzes historical patterns.

Supports:

```
Telemetry trends
Threat trends
Incident trends
Risk trends
Operational trends
```

12.4 Simulation Engine

Performs predictive simulations.

Supports:

```
What-if scenarios
Attack propagation
Mission simulations
Recovery simulations
Capacity simulations
```

12.5 Scenario Engine

Builds alternative outcomes.

Produces:

```
Best-case scenarios
Expected scenarios
Worst-case scenarios
Mission scenarios
Risk scenarios
```

12.6 Confidence Engine

Calculates forecast confidence.

Supports:

```
Prediction confidence
Forecast confidence
Trend confidence
Scenario confidence
Model confidence
```

12.7 Explainability Engine

Provides transparent forecasting explanations.

Produces:

```
Prediction rationale
Evidence references
Historical comparisons
Confidence explanation
Policy references
```

13. Universal Object Model Extensions

13.1 Forecast

```
Forecast:
forecast_id
prediction
confidence
horizon
```

13.2 PredictionModel

```
PredictionModel:
model_id
```

```
version
accuracy
updated_at
```

13.3 Scenario

```
Scenario:
scenario_id
likelihood
impact
description
```

13.4 TrendRecord

```
TrendRecord:
trend_id
category
confidence
timeframe
```

14. Knowledge Graph Integration

Relationships:

```
Telemetry
↓
produces
↓
Trend

Trend
↓
supports
↓
Forecast

Forecast
↓
evaluates
↓
Scenario

Scenario
↓
guides
↓
Decision
```

15. Event Bus Integration

15.1 Forecast Events

```
forecast.generated
forecast.updated
forecast.expired
```

15.2 Prediction Events

```
prediction.completed
prediction.failed
```

15.3 Trend Events

```
trend.detected
trend.updated
```

15.4 Simulation Events

```
scenario.simulated
simulation.completed
```

16. Security Data Lake Integration

Consumes:

```
Historical telemetry
Operational metrics
Detection history
Incident history
Response history
```

17. AI Decision Intelligence Integration

Consumes:

```
Decision history
Recommendation history
Confidence scores
Learning feedback
Reasoning context
```

18. Risk Intelligence Integration

Consumes:

```
Risk scores
Business impact
Mission impact
Risk trends
Threat prioritization
```

19. Compliance Integration

Supports:

```
Forecast auditing
Prediction history
Policy validation
Regulatory reporting
Traceability
```

20. Public APIs

20.1 Forecast API

```
GET /forecasts
POST /forecasts
GET /forecasts/{id}
```

20.2 Prediction API

```
GET /predictions
POST /predictions
```

20.3 Scenario API

```
GET /scenarios
POST /scenarios
```

20.4 Trend API

```
GET /trends
GET /confidence
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ predictive_analytics/
■■■ tests/
■ ■■■ predictive_analytics/
■■■ docs/
■ ■■■ predictive_analytics/
■■■ api/
■ ■■■ predictive_analytics/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Predictive Analytics & Forecasting Engine is the trusted forecasting subsystem responsible for producing explainable, policy-governed predictive intelligence across the AQELYN platform.

Every forecast shall be:

```
Explainable
Evidence-backed
Policy-governed
Risk-aware
Mission-aware
Fully auditable
Traceable
Human-reviewable
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Explainable Forecasting
Least Privilege
Continuous Validation
Policy Enforcement
Secure by Design
Human Oversight
```

22.2 Authorization Model

Supported operational roles:

```
SOC Analyst
Threat Hunter
Mission Owner
Risk Analyst
Compliance Officer
Executive Analyst
Security Administrator
Forecast Administrator
```

All forecasts shall be governed through the AQELYN Policy Engine.

22.3 Forecast Integrity

Forecast records shall maintain:

```
Unique forecast identifier
Forecast horizon
Model version
Evidence references
Confidence score
Explanation
Policy references
Review state
Audit trail
```

Forecast history shall be append-only.

22.4 Simulation Integrity

Scenario and simulation outputs shall maintain:

```
Scenario identifier
Input assumptions
Model version
Simulation timestamp
Confidence level
Impact estimate
Evidence references
Audit record
```

No scenario or simulation shall be considered operationally valid without assumptions and confidence records.

23. Forecast Lifecycle

23.1 Prediction Lifecycle

```
Historical Data
↓
Trend Analysis
↓
Prediction
↓
Forecast Generation
↓
Confidence Assessment
↓
Human Review
↓
Published
```

23.2 Simulation Lifecycle

```
Scenario Created
↓
Simulation Executed
↓
Results Evaluated
```

```
↓
Forecast Updated
```

23.3 Audit Lifecycle

```
Forecast Generated
↓
Evidence Linked
↓
Policy Verified
↓
Audit Stored
```

24. Continuous Forecast Operations

The engine continuously evaluates:

```
Forecast quality
Prediction accuracy
Trend evolution
Scenario effectiveness
Model performance
Mission impact
Risk evolution
Policy compliance
```

25. Performance Requirements

The engine shall support:

```
Low-latency forecasting
Enterprise-scale analytics
Concurrent predictions
Continuous simulations
Continuous operation
High availability
```

26. Scalability Requirements

The engine shall scale to support:

```
Global deployments
Multi-region analytics
Large enterprise environments
Distributed forecasting
Hybrid cloud deployments
Long-term historical analysis
```

27. Audit Requirements

Every forecasting operation shall generate immutable audit records.

Audit events include:

```
Forecast generated
Prediction completed
Trend detected
Scenario simulated
```

```
Confidence calculated
Model evaluated
Policy validation
```

28. Failure Handling

28.1 Prediction Failure

```
Prediction cancelled
Failure recorded
Administrator notified
```

28.2 Simulation Failure

```
Simulation stopped
Retry initiated
Audit generated
```

28.3 Confidence Failure

```
Confidence recalculated
Fallback forecast generated
Audit recorded
```

28.4 Policy Failure

```
Forecast blocked
Policy violation recorded
Human review required
```

29. Testing Strategy

29.1 Unit Testing

Validate:

```
Forecast Engine
Prediction Engine
Trend Analysis Engine
Simulation Engine
Confidence Engine
Explainability Engine
```

29.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Security Data Lake
AI Decision Engine
Risk Intelligence Engine
Mission Engine
Compliance Engine
```

29.3 System Testing

Validate:

```
Forecast generation
Trend analysis
Scenario simulation
Confidence calculation
Forecast publication
Forecast auditing
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Forecast integrity
```

29.5 Regression Testing

Verify IS-001 through IS-020 remain unaffected.

30. Acceptance Criteria

IS-021 is complete when:

```
Forecast Engine implemented
Prediction Engine implemented
Simulation Engine implemented
Trend Analysis Engine implemented
Confidence Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/predictive_analytics/
■■■ tests/predictive_analytics/
■■■ docs/predictive_analytics/
■■■ api/predictive_analytics/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-021 introduces the AQELYN Predictive Analytics & Forecasting Engine, providing enterprise-scale predictive intelligence, explainable forecasting, trend analysis, scenario simulation, confidence estimation, and policy-governed predictive decision support.

Major capabilities include:

```
Predictive Analytics
Threat Forecasting
Trend Analysis
Scenario Simulation
Forecast Confidence
```

```
Explainable Forecasting
Mission Forecasting
Risk Forecasting
Historical Prediction
Policy-aware Forecasting
Executive Forecasting
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

```
Specification ID : IS-021
Title : AQELYN Predictive Analytics & Forecasting Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0021
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
EA-0013 COMPLETE
EA-0014 COMPLETE
EA-0015 COMPLETE
EA-0016 COMPLETE
EA-0017 COMPLETE
EA-0018 COMPLETE
EA-0019 COMPLETE
EA-0020 COMPLETE
IS-021 COMPLETE
EA-0021 READY FOR GENERATION
```

34. EA-0021 Engineering Objective

The objective of IS-021 was to introduce a dedicated Predictive Analytics & Forecasting Engine that enables AQELYN to forecast cyber threats, operational risks, mission impacts, infrastructure degradation, behavior changes, and security posture evolution using historical telemetry, knowledge graph context, AI decision intelligence, and risk intelligence.

The engine extends AQELYN from decision intelligence into proactive forecasting and strategic anticipation.

35. EA-0021 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Forecast generation
Prediction modeling
Trend analysis
Scenario simulation
Confidence assessment
Explainability
Learning support
Knowledge Graph integration
```

```
Security Data Lake integration
AI Decision integration
Risk integration
Mission integration
Event publishing
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Predictive Analytics & Forecasting Engine

Predictive analytics and forecasting responsibilities are implemented as a standalone engine rather than embedded in AI Decision Intelligence, Risk Intelligence, or Threat Detection.

Rationale:

```
Clear separation of prediction from decision support and operational detection.
Independent lifecycle and governance.
Better support for scenario simulation and forecasting horizons.
Improved explainability of forecasts and prediction confidence.
```

36.2 Decision 2 - Forecasts Are Advisory and Governed

Forecasts support decision-making but do not automatically trigger high-impact actions without policy-governed review.

Benefits:

```
Prevents over-automation based on prediction uncertainty.
Allows mission and risk owners to review significant forecasts.
Maintains accountability for forecast-driven decisions.
```

36.3 Decision 3 - Scenario Simulation as First-Class Capability

Scenario and simulation outputs are modeled as governed objects.

Benefits:

```
What-if analysis becomes reproducible.
Mission and risk outcomes can be compared.
Response planning can be evaluated before execution.
```

36.4 Decision 4 - Forecast Explainability Is Mandatory

Every forecast requires a rationale, evidence references, confidence explanation, and model version.

Benefits:

```
Forecasts become auditable.
Analysts can understand prediction drivers.
Executives can evaluate confidence.
Compliance can review forecast-driven decisions.
```

36.5 Decision 5 - Event-Driven Forecast Lifecycle

Forecast, prediction, trend, simulation, confidence, and model events are published through the AQELYN Event Bus.

Examples include:

```
forecast.generated
forecast.updated
forecast.expired
prediction.completed
prediction.failed
trend.detected
trend.updated
scenario.simulated
simulation.completed
confidence.updated
```

This maintains loose coupling between AQELYN engines.

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
Forecast
PredictionModel
Scenario
TrendRecord
```

These extend the Universal Object Model without modifying existing object definitions.

37. Architectural Integration Summary

Engine	Integration
IS-001 Kernel	Runtime lifecycle and service registration
IS-002 Universal Object Model	Forecast, prediction model, scenario, trend objects
IS-003 Event Bus	Forecast, prediction, trend, simulation events
IS-004 Evidence Engine	Evidence references supporting forecast and simulation
IS-005 Knowledge Graph	Entity relationships, trends, forecast impact relationships
IS-006 Trust Engine	Evidence trust and forecast confidence
IS-007 Mission Engine	Mission forecasting and mission impact prediction
IS-008 Workflow Engine	Forecast review, validation, publication workflows
IS-009 Policy Engine	Forecast governance, publication, review policies
IS-010 Compliance Engine	Forecast auditing and regulatory traceability
IS-013 Risk Intelligence Engine	Risk forecasts, risk trends, business impact
IS-014 Threat Intelligence Engine	Threat campaigns, actors, indicators, threat evolution
IS-015 SOC Engine	Incident history and operational forecasting context
IS-017 Threat Detection Engine	Detection history, behavior analytics, anomalies
IS-019 Security Data Lake	Historical telemetry, metrics, detection and response history
IS-020 AI Decision Engine	Recommendation history, learning feedback, reasoning context

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/predictive_analytics/
■■■ tests/predictive_analytics/
■■■ api/predictive_analytics/
■■■ docs/predictive_analytics/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces predictive-analytics-specific security controls:

```
Policy-governed forecasts
Human-reviewable predictions
Mandatory forecast explainability
Evidence-backed forecasting
Forecast confidence scoring
Scenario assumption tracking
Immutable forecast audit trail
Prediction model version tracking
Role-authorized forecast administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Predictive analytics
Threat forecasting
Risk forecasting
Mission forecasting
Behavior prediction
Trend analysis
Anomaly forecasting
Capacity forecasting
Operational forecasting
Confidence estimation
Explainable predictions
Scenario simulation
Forecast auditing
Executive forecasting dashboards
```

41. Risks Identified

Risk	Mitigation
Forecast overconfidence	Confidence scoring and human review
Poor model performance	Model evaluation and continuous validation
Unexplainable prediction	Mandatory Explainability Engine
Scenario misuse	Assumption tracking and audit
Policy-violating forecast use	Policy enforcement and publication controls
Outdated historical data	Security Data Lake refresh and lineage
Forecast bias	Audit, feedback, model evaluation
Excessive trust in advisory outputs	Governance and decision-review workflows

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover forecast engine, prediction engine, simulation engine, trend analysis engine, confidence engine, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

44. Dependencies

Required:

```
EA-0001 through EA-0020
IS-001 through IS-020
```

Enables:

```
IS-022 and subsequent executive intelligence, strategic reporting, board-level governance, and future
forecasting components
```

45. Completion Record

```
Engineering Archive : EA-0021
Implementation Specification : IS-021
Title : AQELYN Predictive Analytics & Forecasting Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue
```

46. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
```

47. Engineering Phase Status

```
Completed Engineering Archives : EA-0001 through EA-0021

Current Status:
EA-0021 COMPLETE

Next Implementation Specification:
IS-022 - AQELYN Executive Intelligence & Strategic Reporting Engine
```

EA-0021 is completed and archived. The engineering workflow is consistent with the project rule:

```
Implementation Specification -> Engineering Archive -> Continue
```

From this point onward, the next engineering artifact is IS-022.

48. Engineering Archive Publication Standard

EA-0021 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/
■ ■■■ example_*.md
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
```

```
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-021-001	Provide predictive forecasting	Sections 8, 12, 23	Complete
FR-021-002	Support trend analysis	Sections 8, 12, 23	Complete
FR-021-003	Support scenario simulation	Sections 8, 12, 23	Complete
FR-021-004	Calculate confidence	Sections 8, 12, 22	Complete
FR-021-005	Support forecast governance	Sections 8, 22, 23	Complete
FR-021-006	Publish forecast events	Sections 8, 15, 36	Complete
NFR-021-001	Explainability	Sections 9, 22, 36	Complete
NFR-021-002	High scalability	Sections 9, 25, 26	Complete
NFR-021-003	Low-latency forecasting	Sections 9, 25	Complete
NFR-021-004	Continuous operation	Sections 9, 24	Complete
NFR-021-005	Auditability	Sections 9, 27, 39	Complete
NFR-021-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-021 Purpose	EA-0021 Objective	Defines why the engine exists
Forecast Engine	FR-021-001	Generates forecasts
Trend Analysis Engine	FR-021-002	Analyzes historical trends
Simulation Engine	FR-021-003	Runs predictive simulations
Scenario Engine	FR-021-003	Produces scenario alternatives
Confidence Engine	FR-021-004	Calculates forecast confidence
Explainability Engine	FR-021-005	Supports governance and audit
Event Publisher	FR-021-006	Publishes forecast and simulation events
Security Data Lake Integration	Historical data source	Supplies telemetry and operational history
AI Decision Integration	Decision context	Supplies recommendations and learning feedback
Risk Intelligence Integration	Risk-aware forecasting	Supplies risk scores and trends
Mission Engine Integration	Mission-aware forecasting	Supplies mission context

| Compliance Integration | Forecast audit | Supports traceability and reporting |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0021

EA-0021 was created to archive completion of IS-021 - AQELYN Predictive Analytics & Forecasting Engine.

The archive records the expansion of AQELYN into predictive analytics and forecasting. IS-021 defines the structure needed to generate forecasts, analyze trends, build prediction models, simulate scenarios, calculate confidence, explain forecasts, integrate with the Security Data Lake and AI Decision Intelligence Engine, and support mission-aware and risk-aware forecasting.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Predictive forecasting must be modeled separately from AI decision intelligence. AI Decision Intelligence recommends decisions based on current context, while Predictive Analytics forecasts likely future states and scenarios.

Governance Note

EA-0021 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Forecast

```
forecast_id: FCST-0001
prediction: increased phishing campaign activity against mission-critical users
confidence: 0.82
horizon: 14_days
evidence:
- evidence://trend-phishing-30d
- evidence://threat-campaign-2001
```

52.2 Example Prediction Model

```
model_id: PM-1001
version: v1.0
accuracy: 0.78
updated_at: 2026-07-07T12:00:00Z
scope: threat_campaign_forecasting
```

52.3 Example Scenario

```
scenario_id: SCN-2001
likelihood: moderate
impact: high
description: If privileged credential exposure continues, mission disruption likelihood increases within 7
days.
```

52.4 Example Forecast Event

```
{
"event_type": "forecast.generated",
"forecast_id": "FCST-0001",
"confidence": 0.82,
"horizon": "14_days",
"source_engine": "aqelyn_predictive_analytics_forecasting_engine"
}
```


53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0021.md
PDF/EA-0021.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_predictive_forecast.md
```

54. Final Archive Statement

EA-0021 is the Engineering Archive for IS-021 - AQELYN Predictive Analytics & Forecasting Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0021 COMPLETE
IS-021 COMPLETE
NEXT: IS-022
```